CrunchBite — Digital Ordering Team: Technology Context

Services Owned by This Team

Service	Responsibility	Technology
Order Service	Cart management, order lifecycle (placed → in-kitchen → ready → collected), MICROS POS sync	Lambda + Aurora PostgreSQL 15
Menu Service	Menu catalogue, daypart scheduling, LTO activation, 86-ing propagation, allergen data	Lambda + DynamoDB + ElastiCache Redis
Payment Service	Adyen session creation, capture, refund, saved card token management	Lambda + Adyen REST + Terminal API
Delivery Gateway	Otter webhook ingestion, order injection into kitchen queue, ETA proxy	Lambda + Otter REST API

Frontends Owned by This Team

Client	Technology	Notes
Mobile App	React Native 0.74, Expo managed workflow	iOS 16+, Android 10+; Zustand for local state, React Query v5 for server state
Web Ordering	Next.js 14 (App Router, RSC)	Deployed to CloudFront + S3; shares cart logic with mobile via shared library
Kiosk App	Electron 30 + React 18	Windows 10 IoT; offline SQLite cache for menu; Jotai for kiosk-specific atom store

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Key Technical Constraints

Menu Service

- DynamoDB (not PostgreSQL) — chosen for low-latency reads and DynamoDB Global Tables for multi-region menu replication
- Redis cache (ElastiCache) with a **60-second TTL** on menu responses; the 86-ing SLA depends on cache invalidation being triggered within 5 seconds of the `item.86d` Kafka event
- Menu data is read-heavy (millions of reads/day) and write-light (a few hundred LTO changes/week)

Order Service

- Aurora PostgreSQL Serverless v2 with connection pooling via RDS Proxy
- Every order write is transactional: order record created + Kafka event produced atomically (using outbox pattern)
- MICROS POS is the **authoritative receipt and tax engine**: Order Service calls MICROS before marking order `COMPLETE`; if MICROS is unavailable, order is queued in SQS and retried with exponential backoff

Payment Service

- Adyen REST API for online (app, web); Adyen Terminal API for in-person (kiosk card readers, drive-through)
- Idempotency key = `orderId`; prevents double-charge on Adyen retries
- No card data stored on CrunchBite infrastructure (SAQ-A scope)
- Refunds trigger a Kafka `payment.refunded` event consumed by the Loyalty Service to reverse any points awarded

Delivery Gateway

- Otter sends webhooks with a 3-second acknowledgment requirement; Lambda function must respond within 2.5 seconds (500 ms buffer)
- Idempotency key = Otter order ID; duplicate webhooks are detected by checking the idempotency table in Redis before writing to Order Service

Kiosk Offline Mode

- Local SQLite database on the kiosk device caches the full menu (updated every sync cycle)
- Orders placed offline are stored in a local queue and submitted to the Order Service when connectivity is restored
- Adyen Terminal API supports offline payment capture for up to **5 minutes** (configurable); offline payments are reconciled when the terminal reconnects

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External Integrations (this team)

Integration	Protocol	Direction	Notes
Oracle MICROS POS	MICROS REST API + TCP socket	Order Service → MICROS	Order confirmation, receipt generation, tax calculation
Adyen	REST + Terminal API	Payment Service ↔ Adyen	All payment processing; SAQ-A scope
Otter	REST webhook (inbound)	Otter → Delivery Gateway	Delivery order injection; 3-second ACK SLA
Infor CloudSuite F&B	REST API (batch)	Menu Service ← Infor	Allergen, nutritional data, recipe updates
Revel Systems	REST API	Kiosk App → Revel	Kiosk software layer wrapping MICROS
Google Maps Platform	REST API	Mobile/Web → Maps	Store locator, delivery radius polygon, ETA

Non-Functional Requirements (this team)

Category	Requirement
Order latency	p95 order placement < 800 ms end-to-end
Menu load	< 300 ms (CDN-cached); cache miss < 800 ms

86-ing propagation	≤ 60 seconds from manager action to all channel updates
Kiosk offline	Accept orders for up to 5 minutes without network; sync on reconnect
Otter ACK	Webhook acknowledged within 3 seconds
Throughput	2.4 M transactions/day; burst capacity for 3× normal during LTO promotions
Availability	99.95% for Order, Payment, Menu services during peak (11 AM–2 PM, 5 PM–8 PM)